

# Alejandro Echeverria

Senior Data and AI Solutions Engineer

[nablaservices@outlook.com](mailto:nablaservices@outlook.com)

<https://linkedin.com/in/cuete>

<https://github.com/cuete>

## Professional Summary

A highly skilled Full Stack Senior Software Engineer, I excel in architecting data flows and AI-powered solutions. Using deep technical knowledge, dynamic project management and communication skills, I specialize in delivering scalable, secure, and compliant solutions that transform ambiguity and raw data into clear, actionable insights for industry leaders and research decision-makers.

## Technical Skills

**AI & ML:** Generative and analytical ML/AI, LLMs, AI agents, multi-agent systems, intelligent automation, design patterns; prompt engineering, prompt optimization, and token cost management (FinOps for AI); model selection and right-sizing strategies; Retrieval-Augmented Generation (RAG); latency optimization and performance benchmarking; model evaluation and A/B testing; Azure OpenAI; AI guardrails and Responsible AI practices.

**Scalability & Performance:** Distributed AI pipeline design; horizontal scaling with Kubernetes; streaming AI workloads (Event Hub); async/parallel pipeline architecture; caching strategies for AI services; Azure AI cost monitoring and budget governance.

**Data & Cloud:** Azure Data Factory, Synapse Analytics, Analysis Services, Data Lake Storage, Event Hub, Cosmos DB, Microsoft Fabric; SQL, NoSQL, Data Lake, Redis, Blob.

**Languages:** C#, Python, JavaScript, TypeScript, R, SQL, PowerShell.

**Infrastructure & DevOps:** Docker, Kubernetes; Azure DevOps, GitHub Actions, Drone; ARM, Terraform (IaC); data structures, algorithms, and performance principles.

**Architecture & Frameworks:** Solutions Architecture; .NET, Node.js, Entity Framework; TCP/IP, HTTP, Zero Trust Network; SDL, Scrum, Agile, DevOps.

**Security & Compliance:** Threat modeling, SAML, OAuth2.0, SSO; GDPR, CCPA, HIPAA; Linux systems administration and internals.

## Soft Skills

**Delivery & Communication:** Technical project management (scoping, coordinating workflows, budgeting, tracking deliverables); collaborative cross-team communication, both technical and non-technical; workflow optimization and KPI-driven delivery.

**Analysis & Insight:** Data-driven business insights; actionable metric creation from large raw data; attention to detail and problem-solving.

**Professional:** Leadership, initiative, motivation, and accountability; security-oriented, with a focus on data privacy and responsible use of AI; adaptability and continuous learning mindset.

## Professional Experience

### Senior Software Engineer (Data and Statistical Modeling) – Gates Foundation

*2025 - Current*

*Seattle, WA*

At the Institute for Disease Modeling (IDM), I design and deliver AI and data solutions that power global health research. I architect data pipelines and AI-driven tools that transform complex epidemiology and simulation data into actionable visualizations and decision-support systems, enabling researchers and stakeholders to interpret findings and drive data-informed decisions in infectious disease eradication. I lead rapid prototyping and production delivery of AI solutions adopted by partner institutions in LMICs, identify gaps in research data and processes, and evaluate and recommend emerging AI technologies to maximize the Foundation's impact in Global Health.

### Senior AI Solutions Consultant – Dura Digital

*2025*

*Seattle, WA*

As Senior AI Solutions Consultant, I architected and delivered AI solutions for enterprise clients. I provided architectural guidance for a regulated Azure AI Foundry deployment at a major Canadian financial institution, including multi-region design across US East 2 and Canada Central/East regions to meet PIPEDA and BCFS data sovereignty requirements. Work included governance controls, zero-trust security, private networking, MuleSoft Flex Gateway integration for Claude model access, and compliance-grade observability. Dura Digital is a global IT consulting and services firm dedicated to delivering business, data, and AI-driven growth solutions.

### Software Engineer II – Microsoft

*2019 – 2025*

*Redmond, WA*

I was responsible for driving the development of high-impact, large-scale enterprise cloud applications from initial business requirements to maintenance and support. I utilized Azure cloud infrastructure, DevOps practices, CI/CD, AI/ML, and data analytics to deliver global, world-class enterprise software products. I

conducted threat modeling and implemented zero-trust strategies to secure critical services and managed incident response for high-priority incidents and outages. I delivered global solutions for consumer and enterprise customer support, including AI Services, Communication Services, APIs, machine learning and intelligent automation, speech recognition and analytics, cognitive tools, and metric processing and analysis while ensuring security and information privacy.

## Software Engineer - Motiv Inc. (Microsoft Contract)

2017 – 2019

*Redmond, WA*

I developed secure, high-throughput, cloud applications at Microsoft's Core Platform Engineering Group. I developed APIs and microservices, identified and mitigated software vulnerabilities, enforced data privacy requirements, security, and compliance, implemented DevOps processes, and hardened resources and applications to meet world-class industry standards.

## Software Engineer - Getty Images

2012 – 2017

*Seattle, WA*

I developed and tested web services and databases handling confidential financial data and the time-sensitive royalty calculation processes. Areas of experience included .NET development, relational databases, message brokers, monitoring and integration platforms, CI/CD, DevOps and Agile methodologies, and on-call engineering support. Getty Images is the global leader in multimedia and royalty services.

## Software Engineer in Test - iSoftStone Inc.

2007 – 2012

*Kirkland, WA*

I designed and executed automated tests for web, mobile, and desktop software products. I developed unit and integration tests for web applications and APIs using automated frameworks and custom scripting. I wrote test plans, researched, and implemented test automation tools and frameworks. I led offshore and on-site testing teams. iSoftStone is a global IT and business services provider for Fortune 100 technology companies.

## Education

### Bachelor of Science in Electronics Engineering

Universidad del Valle de Guatemala

## Certifications

- Agile Project Management, Google
- Azure Security Engineer Associate, Microsoft
- Solutions Architecture, University of Washington

- Telecommunications and Networks, America Movil